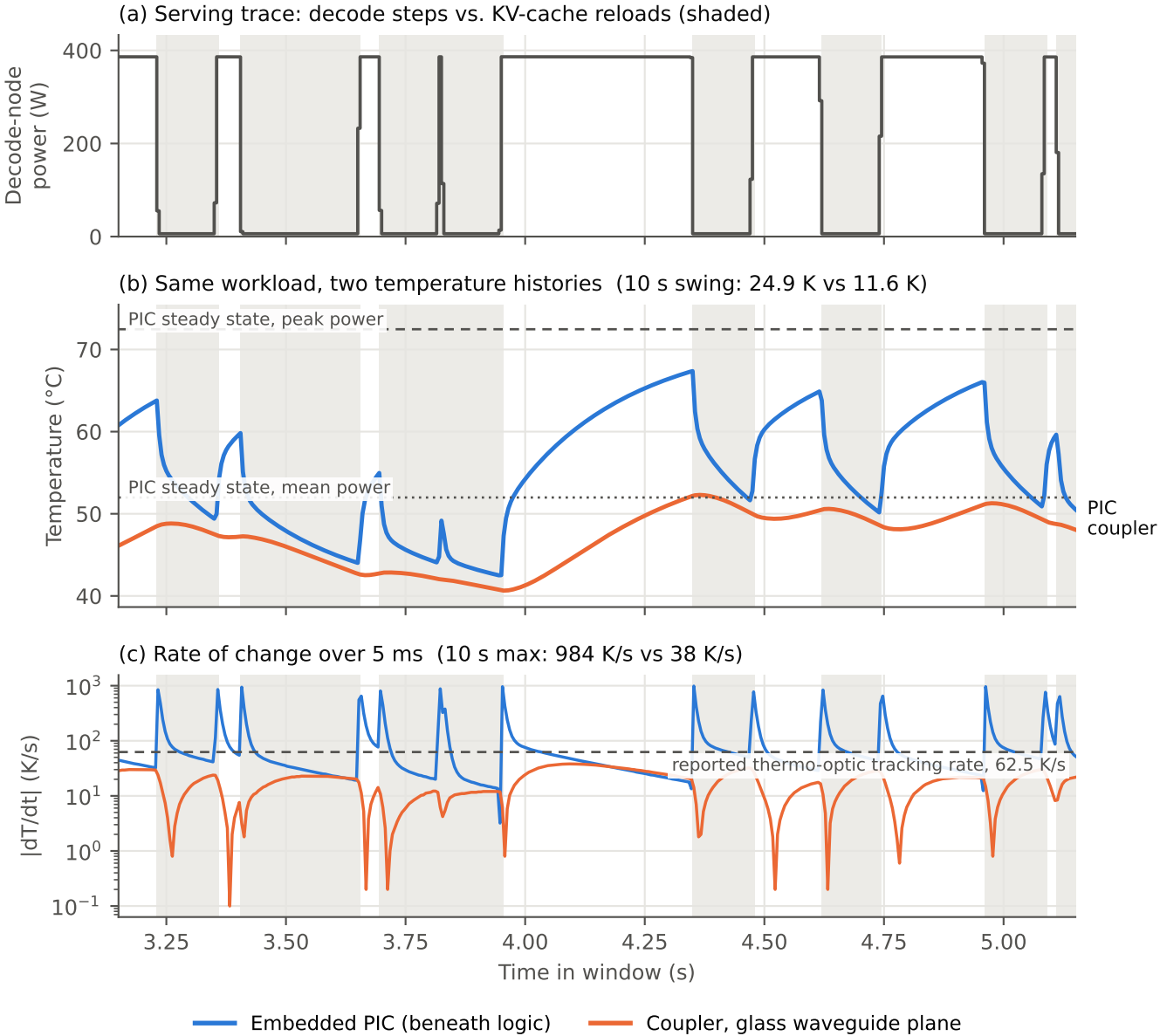
(a) Serving trace: decode steps vs. KV-cache reloads (shaded)
Decode-node power (W)
400
200
0
(b) Same workload, two temperature histories (10 s swing: 24.9 K vs 11.6 K)
PIC steady state, peak power
PIC steady state, mean power
Temperature (°C)
70
60
50
40
PIC
coupler
(c) Rate of change over 5 ms (10 s max: 984 K/s vs 38 K/s)
|dT/dt| (K/s)
10^3
10^2
10^1
10^0
10^{-1}
reported thermo-optic tracking rate, 62.5 K/s
3.25
3.50
3.75
4.00
4.25
4.50
4.75
5.00
Time in window (s)
Embedded PIC (beneath logic)
Coupler, glass waveguide plane